Iterative Graph Propagation for Hyperspectral Anomaly Detection

Jiahui Sheng, Xiaorun Li, and Shuhan Chen, *Member, IEEE*

***Abstract*— As an important task in hyperspectral image (HSI) processing, hyperspectral anomaly detection has gained increasing attention and has been extensively studied. However, most existing methods mostly focus on the statistics spectral distribution of pixels and the spatial correlation of pixels, while ignoring pixels' correlation in the spectral space. Compared with statistics spectral distribution, the spectral correlation represents not only the spectral distribution but also the relations between different pixels in the spectral space. In this article, we propose a novel graph-based hyperspectral anomaly detection method which exploits the correlation of pixels in spectral space. We model the spectral correlation through the structure of the graph which is constructed by a KNN-like strategy. Meanwhile, edge modification (EM) operations including the edge adding (EA) and edge deleting (ED) operations are adopted to further optimize the graph structure. Once the graph structure is determined, the vertices with fewer edges connected will be masked iteratively. This is because vertices with fewer connections have a higher probability of being anomalous compared to others. Then, all the masked vertices will be reconstructed using an iterative graph propagation updating strategy. Since background pixels exhibit higher correlation, they can be effectively reconstructed through graph propagation. However, anomalies are more challenging to reconstruct due to their low spectral correlation. Experiments are conducted on four real HSI datasets which demonstrate the superior performance of the proposed iterative graph propagation anomaly detector (IGPAD) method in detecting small-size anomalies compared with other graph-related and state-of-the-art anomaly detection methods.**

***Index Terms*— Graph propagation, graph-based, hyperspectral anomaly detection, mask.**

I. Introduction

DIFFERENT from visible light and infrared images, hyperspectral image (HSI) possesses additional spectral dimension that provides detailed spectral information of ground objects [1], [2]. Due to the powerful ability to acquire spectral information of ground objects, HSIs can be used in various fields such as target detection, unmixing, and anomaly detection [3], [4]. Among these fields, anomaly detection is one of the hotspots, which is usually regarded as an unsupervised target detection without prior spectral information of target [5], [6]. The main purpose of anomaly detection is to detect the unusual objects that possess distinct spectral signatures from the dominant background. In practice, the prior spectral information of target is hard to acquire that makes anomaly detection much more practical.

Due to the lack of prior spectral information about anomaly targets, achieving high-performance hyperspectral anomaly detection becomes a challenging task while the determination of anomaly becomes ambiguous [7]. Generally, researchers consider objects that account for a smaller portion of the HSI, along with large differences in spectra, to be anomalies. To keep consistent with the definition of anomaly and avoid controversy, the anomaly mentioned in the article refers to small-size anomaly.

With extensive research efforts by scholars in the field of hyperspectral anomaly detection, a considerable number of methods were proposed over the past two decades. These existing methods can be roughly classified into four categories.

1) Statistics-based methods, e.g., Reed-Xiaoli anomaly detector (RXAD) [8] and its variations [9], [10], [11], [12].
2) Representation-based methods, e.g., low-rank and sparse-based methods [13], [14], [15], [16] and collaborative representation-based methods [17], [18], [19], [20], [21].
3) Decomposition-based methods, e.g., matrix decomposition-based methods [22], [23], [24], [25] and tensor decomposition-based methods [26], [27], [28], [29].
4) Deep learning-based methods, e.g., residual-based methods [30], [31], [32], [33], [34] and feature-based methods [35], [36], [37].

Although a diverse range of approaches have been proposed for hyperspectral anomaly detection, they most focus on the statistics spectral distribution of pixels and the spatial correlation of pixels while ignoring spectral correlation of pixels. The statistics-based methods like RX [8] utilize differences in the statistics spectral distribution between the anomalous and the background pixels. Spatial correlation of pixels is always described as the smoothness and similarity in local spatial area. For example, Gao et al. [38] exploit the information of pixel's spatial neighbors to reconstruct the pixel itself. Fan et al. [32] embed a superpixel segmentation-based graph regularization term into AE which preserves the geometric structure and the local spatial consistency. Compared with statistics spectral distribution, the spectral correlation represents not only the spectral distribution but also the relations between different pixels in the spectral space. Moreover, by utilizing the spectral correlation of pixels, we can distinguish the

Manuscript received 22 May 2024; revised 18 June 2024; accepted 15 July 2024. Date of publication 17 July 2024; date of current version 30 July 2024. This work was supported in part by Zhejiang Province Nature Science Foundation of China under Grant LQ24F030004 and in part by the Joint Fund Project of Chinese Ministry of Education under Grant 8091B022118. *(Corresponding author: Xiaorun Li.)*

The authors are with the College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: jiahuisheng@zju.edu.cn; lxr@zju.edu.cn; 11410057@zju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3430111

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

out-of-distribution background pixels from the out-of-distribution anomaly pixels, since they possibly possess different spectral correlations. These out-of-distribution background pixels are often incorrectly categorized as anomalies in existing methods.

To leverage the spectral correlation, several graph-based methods have been proposed. In [39], the weighted vertex volume (WVV) metric, which depicts the connectivity of a given vertex to the rest of the graph, is used to measure the anomalousness of each pixel in the HSI. In [40], spectral graph theory is adopted to model the background for anomaly detection. Based on graph pixel extraction, a hyperspectral anomaly detection method named local graph anomaly detector (LGAD) was proposed which constructs a vertex- and edge-weighted graph [41]. Similar in essence to LGAD, a graph-based method referred to as graph evolution-based vertex extraction (GEVE) is proposed, considering that the relations between pixels evolve dynamically at different scales [42]. In addition to methods that are primarily based on graph, some researchers have combined graph theory with other models. Cheng and Wang [43] proposed a low-rank representation model [graph and total variation regularized low-rank representation (GTVLRR)] that exploits the graph and total variation regularization. Similarly, Fan et al. [32] embedded a superpixel segmentation-based graph regularization term into an autoencoder model, which helps preserve the geometric structure and the local spatial consistency of HSI. Furthermore, in other fields of HSI, graph propagation has been utilized for information dissemination. Rolland et al. [44] generalize the heat diffusion equation to a graph-based diffusion equation and achieve tensor completion. In addition, Zhang et al. [45] propose the dual sparse representation graph-based collaborative propagation (DSRG-CP) for HSI classification which propagates labels on two graphs iteratively.

While many excellent graph-based approaches have been proposed for hyperspectral anomaly detection, there is a lack of work that effectively utilizes the latent information hidden in the structure of the spectral correlation graph. In some previous methods, the graph has been used primarily as auxiliary information to preserve the local geometric structure in new space [32], [43], without deeply exploring the inherent properties of the graph itself. In other graph-based methods [39], [41], [42], the spectral relation of pixels has been considered to construct the graph and realize anomaly detection using the fragile relation of anomaly pixels with other pixels. However, the existing graph-based methods only construct graph with the spectral similarity and further define graph-based similarity metrics for distinguishing background and anomalous pixels, rather than extensively exploring the latent spectral correlation represented by the specific graph structure. For real-world scenarios where there are challenges such as spectral variability or mixed spectra, solely relying on spectral similarity to identify background pixels can lead to a decreased separability between anomalies and the background, resulting in increased false alarms.

To address this gap, in this article, we propose a novel graph-based method called iterative graph propagation anomaly detector (IGPAD) that effectively leverages the correlation of pixels in spectral space to distinguish anomaly from background through an iterative graph propagation updating strategy. This article proposes that from the perspective of spectral correlations, although the background pixels in the aforementioned scenarios may not be sufficiently similar, they do possess spectral correlations. Therefore, this work introduces the means of information propagation to exploit these spectral correlations which is represented by the graph structure. By solely leveraging the information hidden in the spectral correlation graph, our proposed method has acquired promising performance for hyperspectral anomaly detection.

The IGPAD successfully constructs a graph to model the correlation of pixels in spectral space, and the information can propagate through the construction of the graph. Specifically, the IGPAD employs a mutual K-nearest-neighborhood strategy to obtain the initial graph structure. Subsequently, we will impose edge modification (EM) operations on the graph structure by deleting weak edges and adding extra edges. Once the graph structure is established, the vertices with fewer connected edges will be iteratively masked. These vertices are more likely to correspond to anomalies and are prioritized for masking. To leverage the information embedded in the structure of the graph, an iterative graph propagation updating strategy is adopted to recover the value of the masked vertices. Due to the strong correlation among background pixels, masked background pixels have a higher probability of being effectively recovered, while masked anomaly pixels cannot. Finally, the reconstruction error will be utilized for anomaly detection. To be specific, the main contributions of this article can be summarized as follows.

1) We propose a novel graph-based method called IGPAD that effectively leverages the pixelwise spectral correlation for hyperspectral anomaly detection. This method successfully models the spectral correlation of pixels through the graph structure and proposes a new graph-based framework for hyperspectral anomaly detection. In this framework, we deeply exploit the latent information of graph structure through the propagation of information.
2) A mutual-KNN-based graph construction method and spatial distance-based edge weight are employed to construct the initial graph. To further optimize the graph structure and improve the robustness of IGPAD, edge modification operations, including edge adding (EA) and edge deleting (ED) operations, are adopted, which make the IGPAD more practical.
3) We adopt an iterative mask (IM) strategy to ensure that anomaly vertices are prioritized for masking. For instance, a vertex will still be masked even if it has numerous edges connected to other vertices, as long as its most adjacent vertices are identified as suspicious or potentially anomalous vertices.
4) The iterative graph propagation update strategy is exploited to reconstruct the masked vertices in the graph. The Laplacian of the graph is utilized to implement weighted average graph propagation, enabling adaptive adjustment of the propagation speed based on the anomalous degree.

The rest of this article is organized as follows. Section II provides an introduction to preliminary knowledge.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

In Section III, we present a detailed description of our proposed IGPAD, including the construction of the graph, edge modification operations, iterative mask strategy, and iterative graph propagation update strategy. Section IV discusses experimental results and provides comparisons with graph-related and state-of-the-art algorithms. Finally, Section V concludes this article.

II. Preliminaries

In this section, preliminary knowledge of the proposed method will be introduced.

A. Laplacian of a Graph

A graph is made up of vertices (nodes) which are connected by edges (links). Depending on whether the edge has direction, graph can be divided into undirected graph and directed graph. The graphs in this article are by default undirected graphs if not specified.

Supposing G is a graph with vertex set V and edge set E consisting of unordered pairs of V, graph G can be termed as $G = \{V, E\}$. Generally, a graph is represented by an adjacency matrix A, whose elements indicate whether pairs of vertices are adjacent or not. Considering an unweighted graph that containing N vertices, the adjacency matrix can be defined as follows:

$$A_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in E \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $(v_i, v_j) \in E$ means that v_i and v_j are adjacent and $A_{ij} = 1$ means that edges are unweighted in graph G which can be replaced with other nonnegative number if the edges have weights. The special note is that $A_{ij} = 0$ if $i = j$ which means that there is no loop in graphs used in this article.

In a graph G, let D denote the degree matrix of G which is a diagonal matrix with $D_{ii} = \sum_j A_{ij}$. The Laplacian of graph G, denoted by L, is defined to be $D - A$. To be specific, the elements of L are given by

$$L_{ij} = D_{ij} - A_{ij} = \begin{cases} d_i, & \text{if } i = j \\ -1, & \text{if vertex } i, j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where d_i represents the degree of vertex i. L, in fact, describes the smoothness of the value of the vertices in graph. Aiming at balancing the influence of different vertices, normalization is adopted to Laplacian matrix, including symmetrically normalized Laplacian and random-walk normalized Laplacian. Among them, the random-walk normalized Laplacian is defined as $L^{\text{rw}} = D^{+}L$ where D^{+} represents Moore–Penrose inverse. The elements of L^{rw} are given by

$$L_{ij}^{\text{rw}} = \begin{cases} 1, & \text{if } i = j \text{ and } d_i \neq 0 \\ -\dfrac{1}{d_i}, & \text{if vertex } i, j \text{ are adjacent} \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

III. Proposed Method

This section presents an introduction to our proposed method (IGPAD). As shown in Fig. 1, our IGPAD method includes four steps: 1) mutual-KNN strategy for graph construction; 2) edge modification, including EA and ED operations; 3) iterative mask strategy; 4) iterative graph propagation update strategy.

A. Mutual-KNN Strategy for Graph Construction

In the part of graph construction, we first build a mutual-KNN graph which depicts the spectral correlation of pixels. Then, the spatial distance-based edge weight method will be adopted to acquire the weights of edges.

1) Mutual-KNN Graph: We use a mutual-KNN strategy to construct graph in this method. Suppose there is an HSI size of $H \times W \times B$, which means that there are $N = H \times W$ pixels with B bands, we take the corresponding spectra as the signature of each pixel (vertex of graph). Depending on the signatures of vertices, we can get K-nearest neighbors of each vertices. If two vertices are the K-nearest neighbors of each other (having high spectral similarity), we will connect them, which means that these two vertices are adjacent in graph. By utilizing the similarity of the spectra as the foundation for constructing the graph, we are able to model the correlation of pixels in the spectral space with graph. In general, the background vertices tend to have a significantly higher number of connected edges compared to the anomaly vertices. This is due to the homogeneous and dominant nature of the background in HSI, which is usually depicted as the low-rank property of HSI. The correlation-based graph construction is illustrated by Fig. 2.

2) Spatial Distance-Based Edge Weight: Although the mutual-KNN graph can model the spectral correlation of pixels, the graph ignores the spatial information of HSI which contains a lot of useful information for hyperspectral anomaly detection. Considering the spatial distance between vertices, we adopt the spatial distance-based edge weight rather than just using 1 as the weight of edges. The spatial distance-based edge weight is formulated as

$$w_{ij} = \frac{1}{1 + e^{d_{ij} - \sigma}} + 1 \tag{4}$$

where w_{ij} represents the weight of the edge between vertex i and vertex j, d_{ij} represents the Euclidean distance between vertex i and vertex j in geometric space, and σ is the local window threshold. Using the spatial distance-based edge weight, (1) and (3) are updated as follows:

$$A_{ij} = \begin{cases} w_{ij}, & \text{if } (v_i, v_j) \in E \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

$$L_{ij}^{\text{rw}} = \begin{cases} 1, & \text{if } i = j \text{ and } d_i \neq 0 \\ -\dfrac{w_{ij}}{d_i}, & \text{if vertex } i, j \text{ are adjacent} \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

B. Edge Modification

After the step of graph construction, we have acquired a mutual-KNN graph with spatial distance-based edge weights.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

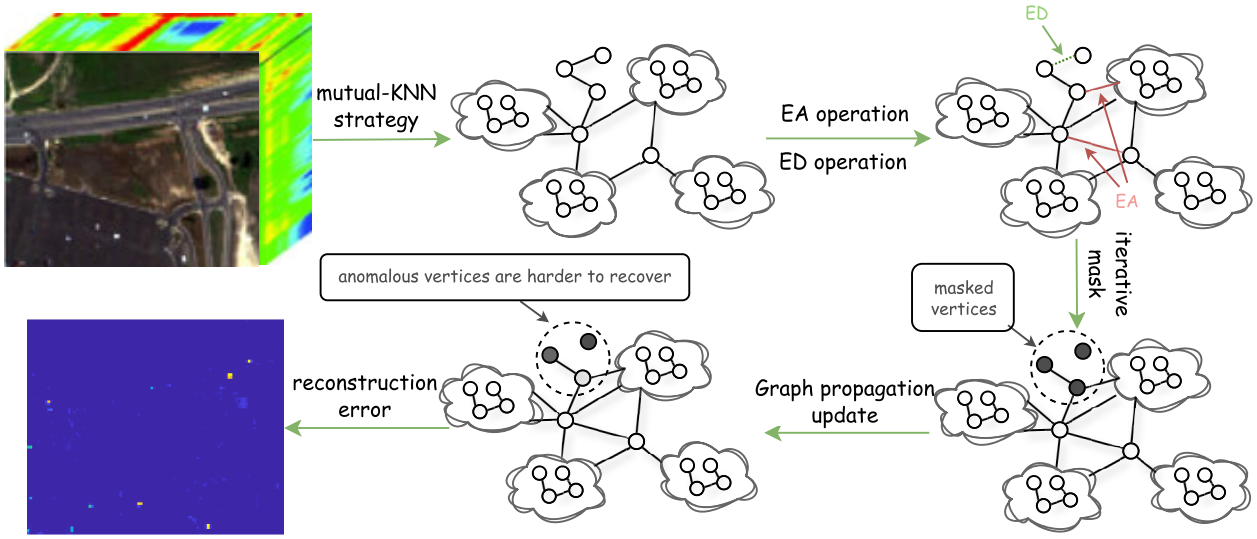


Fig. 1. Overview of the proposed IGPAD method in this article. At first, the mutual-KNN strategy is used to construct the initial spectral correlation graph. Subsequently, EA and ED operations are adopted to modify the graph to deal with the missing and spare edges. Then, the iterative mask strategy and iterative graph propagation update will be used sequentially to reconstruct vertices by latent information of graph structure. Finally, we take the reconstruction error as the detection map.

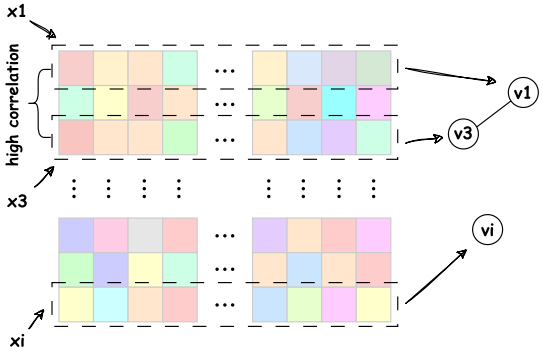


Fig. 2. Correlation-based graph construction. x_i here represents the ith pixel in HSI and corresponds to vertex v_i in the graph. As illustrated, x_1 and x_3 have similar spectra which means that they have high spectral correlation so that v_1 and v_3 are connected in the graph.

However, merely exploiting mutual-KNN strategy to construct graph is not enough. Therefore, two edge operations, EA and ED, are adopted to modify the graph.

1) EA Operation: In certain cases, it is possible that there are two vertices whose signatures are extremely similar while they are not the mutual K-nearest neighbors, since one of them is very popular among other vertices in the graph, potentially serving as the center of a cluster. Therefore, the vertex similar to the center vertex has a probability of being missed by mutual-KNN strategy and isolated by its neighbors, which is considered as the pseudoisolated vertex. In this situation, despite not being mutual K-nearest neighbors, the two vertices should still be connected, since their strong spectral similarity suggests a close correlation between them that should be captured by the graph. To be specific, we will first categorize all the vertices into suspicious vertices and suspectless vertices based on their adjacency number through the iterative mask strategy (which will be described later). Once the vertices are categorized, our focus will be solely on the suspicious vertices, as they are the ones that potentially contain anomalies. We will then examine each suspicious vertex to determine if it has any missing neighbors among the suspectless vertices, using a threshold. The threshold is defined by (7) which is adaptive to the datasets. If a suspicious vertex has missing neighbors within the suspectless vertices, indicating a potential connection that was not initially established, we will connect the vertex to its missing neighbors

$$\sigma_{\mathrm{EA}} = d_1 \times \sqrt{C} \tag{7}$$

where C represents the number of channels and d_1 represents EA coefficient, ranging from 0 to 1.

2) ED Operation: In addition to the previously mentioned scenario of missing edges, there are situations where vertices are connected not due to high correlation but because they are isolated or "lonely" in the spectral space. In other words, when the signatures of certain vertices are distinct from the majority of vertices, there is a possibility that isolated vertices may become mutual K-nearest neighbors, even if they are dissimilar to each other. This occurs because these isolated vertices form a distinct subgroup, leading to them becoming each other's closest neighbors. To deal with this problem, we impose the ED operation on the graph. Specifically, we will examine the edges between suspicious vertices, and each edge is evaluated to determine whether the similarity, measured by the spectral Euclidean distance, between the two connected vertices surpasses a threshold. The threshold for ED is defined by (8) which is adaptive to the datasets. If the similarity between the two connected vertices is below the threshold, indicating a low level of correlation or similarity, that corresponding edge will be deleted from the graph

$$\sigma_{\mathrm{ED}} = d_2 \times \sqrt{C} \tag{8}$$

where d_2 represents ED coefficient, ranging from 0 to 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

By incorporating these operations, the graph is enhanced to better represent the spectral correlation among pixels, resembling the concept of "closeness" in social networks.

C. Iterative Mask Strategy

The goal of the mutual-KNN strategy for graph construction and edge modification is to create a graph that accurately represents the spectral correlation among pixels in HSI. After obtaining the graph, an iterative mask strategy will be adopted to mask certain vertices that have a higher probability of being anomalies. As mentioned above, due to the homogeneity and majority of the background, background vertices generally have a higher number of connected edges compared to anomaly vertices. Therefore, vertices with fewer edges, referred to as suspicious vertices, are more likely to correspond to anomalies in HSI. Instead of directly masking the suspicious vertices in a single step, we employ an iterative mask strategy to mask them iteratively. This iterative strategy enhances the robustness of the method.

Like the internal vertices in Fig. 3, the anomaly vertices can possess a lot of edges while most of their adjacent vertices are suspicious vertices. In this situation, we have a low confidence that these vertices represent the background. To address this scenario, we propose the iterative mask strategy. This strategy involves initially masking the most suspicious vertices, which are the vertices with the least number of edges in the graph, and deleting the edges connected to the masked vertices. Then, we expand the range of suspicious vertices and perform the same operation on the remaining graph. This process is repeated until completion. The specific workflow is shown in Fig. 3, which provides an example of iteratively masking suspicious vertices whose edges are no more than 4.

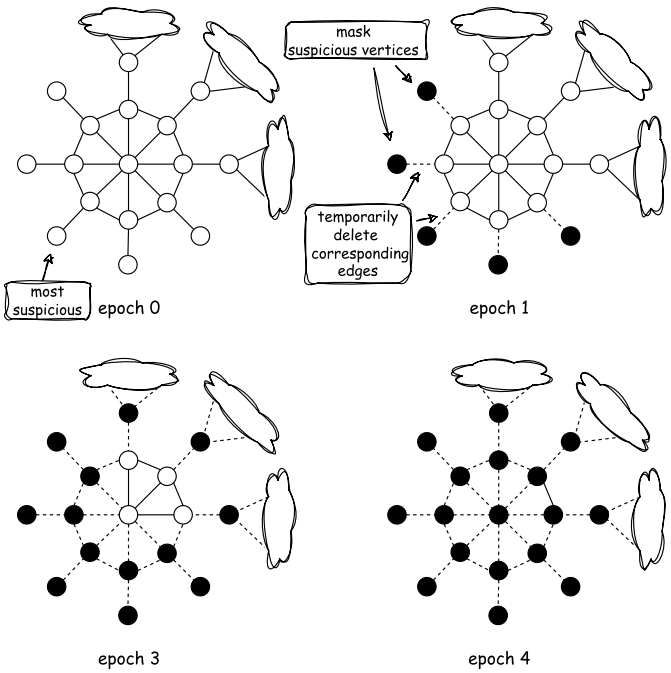


Fig. 3. Iterative mask strategy. This iteratively masks suspicious vertices whose edges are no more than 4. Epoch 1 masks outermost vertices whose edges are no more than 1 and temporarily deletes their corresponding edges. Epoch 2 is skipped here. Epoch 3 masks vertices whose edges are no more than 3 after epochs 1 and 2 and deletes corresponding edges. Epoch 4 masks vertices whose edges are no more than 4 after epoch 3 and deletes corresponding edges. The cloud here represents the suspectless vertices which would not be masked. What need to be emphasized is that the edges here are deleted temporarily for mask, and the edges are retained in the graph during the recovery process.

D. Iterative Graph Propagation

After masking the suspicious vertices, the next step is to recover the masked vertices. The task of our method is still a reconstruction task, and the objective function is formulated as

$$\min_{\mathbf{V}} \|\tilde{\mathbf{V}} - \mathbf{V}\|_F, \quad \text{s.t.} \mathcal{S}(\mathbf{V}) \tag{9}$$

where $\mathbf{V} \in N \times B$ is the matrix that represents all vertices after reconstruction and $\tilde{\mathbf{V}}$ is the real matrix of vertices. $\|\cdot\|_F$ represents the Frobenius norm, and $\mathcal{S}(\mathbf{V})$ represents that $\mathbf{V}$ should be consistent with the structure of graph.

Rather than relying on algebraic methods to solve (9), we leverage the latent information present in the graph structure to reconstruct the masked vertices. Naturally, we think of exploiting the retained vertices (unmasked vertices) to propagate information along the edges, following certain propagation rules. One commonly used propagation rule is the weighted average propagation rule. According to this rule, each masked vertex performs a weighted average update based on the information obtained from its adjacent vertices. The rule is depicted as follows:

$$\mathbf{v}_i = \frac{\sum w_j \mathbf{v}_j}{\sum w_j} \tag{10}$$

where $\mathbf{v}_j \in 1 \times B$ is the adjacent vertex of the masked vertex $\mathbf{v}_i$ and w_j is the propagation weight of $\mathbf{v}_j$.

In our graph, it is not feasible to update all the masked vertices in a single step due to the potential connections between masked vertices, as shown in Fig. 3. Therefore, we adopt an iterative graph propagation strategy that gradually updates the masked vertices by weighted average to propagate information from the unmasked and recovered vertices as illustrated in Fig. 4.

To enable iterative update, the weighted average propagation rule is formulated as follows:

$$\begin{aligned} \mathbf{v}_i^{k+1} &= \mathbf{v}_i^k + \alpha\left(\frac{\sum w_j \mathbf{v}_j}{\sum w_j} - \mathbf{v}_i^k\right) \\ &= \mathbf{v}_i^k - \alpha\frac{\mathbf{v}_i \sum w_j - \sum w_j \mathbf{v}_j}{\sum w_j} \\ &= \mathbf{v}_i^k - \alpha\frac{\mathbf{v}_i d_i - \sum w_j \mathbf{v}_j}{d_i} \end{aligned} \tag{11}$$

where $\alpha > 0$ is used to determine the update step size.

To further simplify the formula, we can express the second part of the equation in terms of the random-walk normalized Laplacian of the graph. In this approach, we update all masked vertices simultaneously, as shown below

$$\mathbf{V}^{k+1} = \mathbf{V}^k - \alpha \mathbf{M}\mathbf{L}^{\text{rw}}\mathbf{V}^k \tag{12}$$

where $\mathbf{V}$ is a matrix that represents all vertices in graph G and $\mathbf{L}^{\text{rw}}$ stands for the random-walk normalized Laplacian matrix

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

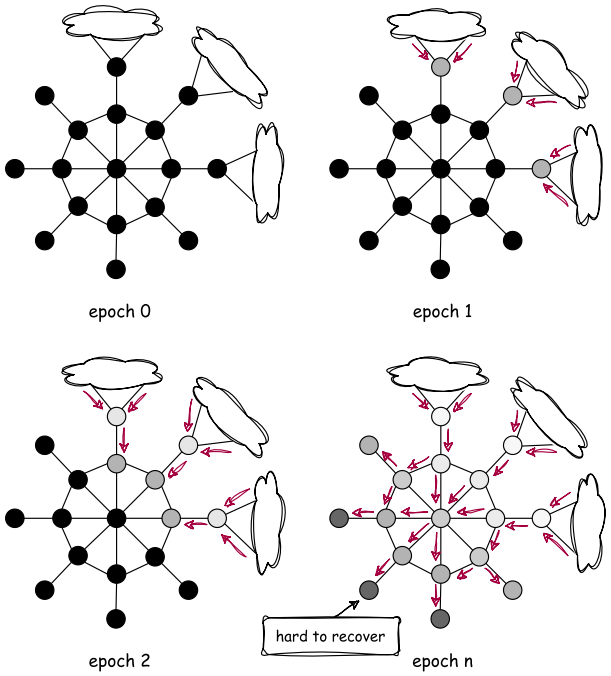


Fig. 4. Overview of iterative graph propagation. The upper right vertices are connected to other unmasked vertices, which are represented as clouds in the figure. The red arrows represent the direction of information flow, and the shade of the vertices represents the degree of recovery, with lighter shades indicating better recovery. As shown in the figure, the information is propagated to the further masked vertices, and the masked vertices gradually recover as the number of epochs increases. However, some vertices are still difficult to recover, such as the vertices in the lower left corner, which are far from the unmasked vertices and likely represent anomalies.

of graph G. The mask matrix $\mathbf{M}$ is a diagonal matrix whose ith element is 1 if $\mathbf{v}_i$ is masked and will be 0 if not. Mask matrix is used here to make sure that the unmasked vertices would not be modified during the graph propagation.

Although the anomaly vertices, which have low correlation with other vertices, are less likely to be recovered through graph propagation, there is still a possibility that they can be recovered as the iterations' progress. To prevent the recovery of anomalies, we introduce a vertex-adaptive update step size to replace the original vertex-consistent update step size α. This adjustment slows down the graph propagation on the anomaly vertices. As mentioned earlier, the vertex degree represents the anomalousness of a vertex to some extent. Therefore, we can directly utilize the degree matrix of the graph to construct the vertex-adaptive update step size, which is formulated as follows:

$$\begin{aligned}\mathbf{V}^{k+1} &= \mathbf{V}^k - \alpha\mathbf{M}\mathbf{D}\mathbf{L}^{\mathrm{rw}}\mathbf{V}^k \\ &= \mathbf{V}^k - \alpha\mathbf{M}\mathbf{L}\mathbf{V}^k. \end{aligned} \tag{13}$$

Through iterative graph propagation, the partial background vertices that are considered as suspicious vertices and masked will be recovered well while the anomaly vertices that are masked would not. After convergence, the reconstruction error between reconstructed HSI and real HSI will be computed and used as the detection map.

TABLE I
SPECIFIC DESCRIPTION OF FOUR DATASETS

Dataset	The number of anomalies	The ratio of anomalies	Size (H×W×C)
HYDICE Urban	21	0.26%	80×100×174
Pavia	68	0.30%	150×150×102
Hyperion	32	0.32%	100×100×145
abu-beach-1	19	0.08%	150×150×188

IV. EXPERIMENTS AND ANALYSIS

In this section, we first provide specific descriptions of the datasets. Subsequently, we present the baselines used for comparison and the metrics for quantitative analysis. Then, we conduct a comprehensive parameter analysis to explore the influence of the significant parameters. Moreover, in order to demonstrate the validity of the individual components of the proposed IGPAD detector, we give a sufficient ablation study. Finally, both the detection map and the quantitative metric of our proposed IGPAD and baseline methods are presented for a better comparison, which prove the superiority of our proposed IGPAD method. The components of this section are specified as follows.

A. Dataset

To evaluate the performance of our proposed IGPAD detector, four datasets, including HYDICE Urban, Pavia, Hyperion, and abu-beach-1, are used in experiments. The specific description of four datasets is displayed in Table I, including the number of anomalies, the ratio of anomalies, and the HSI size. The pseudocolor image, ground-truth map, and mean spectrums of background and anomalies are shown in Fig. 5.

Through Table I and Fig. 5, you can find that the anomalies in the selected datasets have a low probability and small proportion in his, since the proposed method is based on the low spectral correlation property of anomalies which is not satisfied by large-size anomaly. Meanwhile, the object that has a large size is considered as anomaly uncertainly as discussed in [7].

B. Experimental Setup

1) Baselines: Eight contrast methods, consisting of classical, graph-related, and state-of-the-art methods, are used in this experiment to verify the superiority of the proposed IGPAD method. The elaborately selected methods for comparing are briefly depicted as follows.

1) *GRX [8]:* The classical statistics-based method which is based on the assumption that the background satisfies the Gaussian distribution.
2) *CRD [17]:* The collaborative representation detector which is based on the philosophy that background can be represented collaboratively.
3) *GTVLRR [43]:* A representation-based method which embeds the graph regularization into the low-rank representation model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

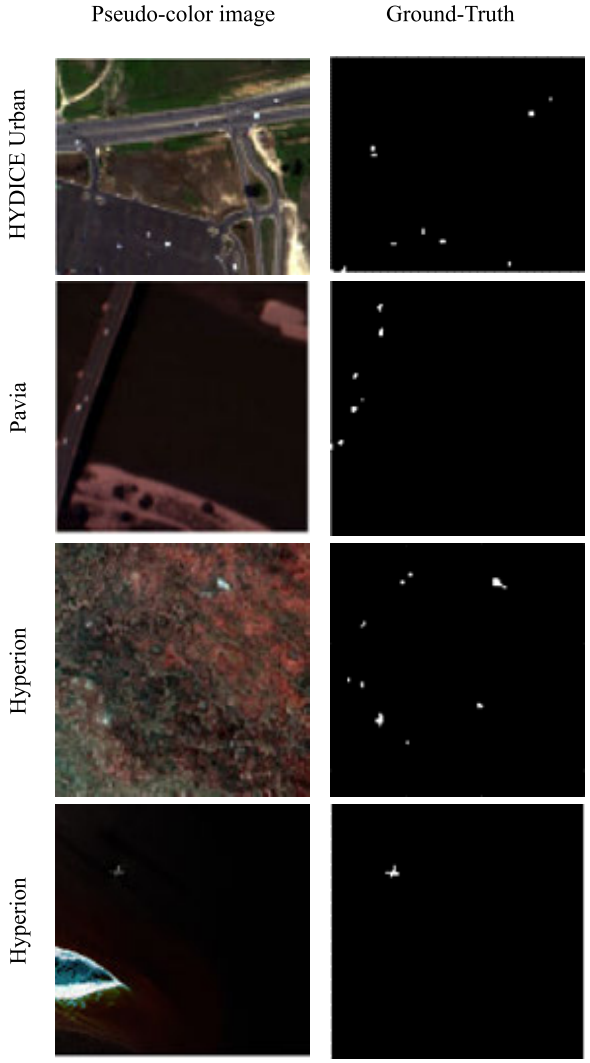


Fig. 5. From left to right are pseudocolor images and ground truth of four datasets.

4) *LSMAD [22]:* A decomposition-based method which decomposes the HSI into low-rank component and sparse component.
5) *PTA [29]:* A decomposition-based method proposing a prior-based tensor approximation (PTA) for tensor decomposition.
6) *RGAE [32]:* A deep learning-based method which combines graph theory with autoencoder.
7) *Auto-AD [31]:* A typical deep learning-based method, one of the state-of-the-art methods.
8) *SSG [46]:* A graph-based method which takes both spatial and spectral information into consideration and constructs spectral–spatial graph.

2) Metrics: As a type of classification task, hyperspectral anomaly detection regularly takes area under the curve (AUC) of receiver operating characteristic (ROC), which is widely used as a metric for binary classification task, as an evaluation metric of the result of the detection. However, although AUC of ROC is widely used in hyperspectral anomaly detection, it has the disadvantage of being falsely high as well as not being able to match well with the detection result. Considering the drawback of ROC, Chang [47] introduced 3-D ROC to hyperspectral anomaly detection which split performance of hyperspectral anomaly detection into target detectability and background suppression. The related metrics of 3-D ROC curve consist of $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{SNPR}, $AUC_{TD\text{-}BS}$, and AUC_{ODP}.

In this article, we take $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$ as our main metrics to quantize the overall performance, target detectability, and background suppression, respectively.

3) Implementation Details: To demonstrate the robustness of the proposed IGPAD method across different datasets, we set all the same parameters on four used datasets. To be specific, we set the parameters K of KNN, the mask number m, σ in distance-based edge weigh, EA coefficient d_1, ED coefficient d_2, and update step-size α to 100, 30, 15, 0.02, 0.06, and 0.03, respectively. Additionally, we conduct all the following experiments on a Windows 11 computer with 32-GB memory, 13th Gen Intel[1] Core[2] i7-13700K CPU, and NVIDIA GeForce RTX 4060Ti GPU.

C. Parameter Analysis

1) Parameter K in Mutual-KNN Strategy: In the IGPAD method, one of the most important manual-set parameters is the parameter K in mutual-KNN graph construction. To figure out the influence of parameter K on the performance of the proposed method, we conduct an analysis of parameter K by comparing the performance under different K values. All parameters are fixed except for the parameter K; the mask number is taken as 30. As shown in Fig. 6, the value of $AUC_{(D,F)}$ on four datasets is fairly high when the parameter K is near 100 which demonstrates the robustness of parameter K on different datasets. On Pavia and abu-beach-1 datasets, the value of $AUC_{(D,F)}$ has a sudden decline when the parameter K reaches a high value, since the adjacency number of each vertex will increase as the parameter K grows larger, and the prime mask number m will not be sufficient to mask enough number as well as all potential anomalous vertices.

2) Mask Number m: Another decisive factor that influences performance is the parameter m, which represents the number of masked vertices and directly affects the quality of the mask. The parameter m should be sufficiently large to ensure that every anomaly vertex is masked during the masking stage. The parameter K is set to 100 in this part. The specific influence is conveyed through Fig. 7, which exhibits the generalization of parameter m across different datasets. Similar to the parameter K, there is a big drop in performance when the value of m is relatively small which is caused by anomalous vertices leak during masking (in other words, some anomalous vertices are not masked in the stage of masking).

D. Ablation Study

In this section, we conduct an ablation study to evaluate the contribution of different modules in the IGPAD method, including edge modification operations and iterative mask

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

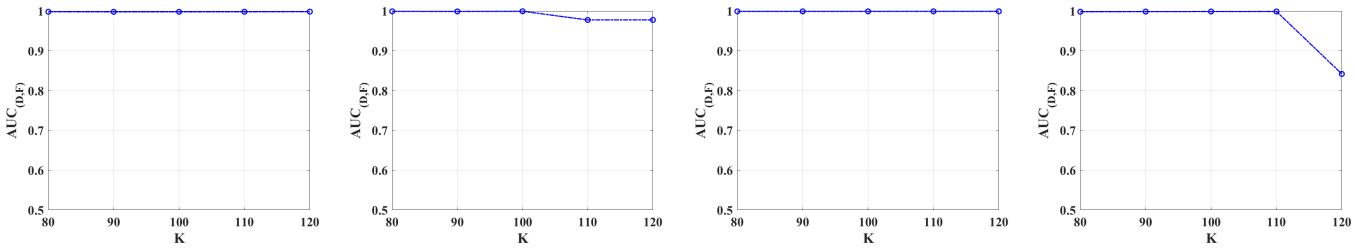


Fig. 6. Parameter analysis of K on four datasets. (a) HYDICE Urban. (b) Pavia. (c) Hyperion. (d) Abu-beach-1.

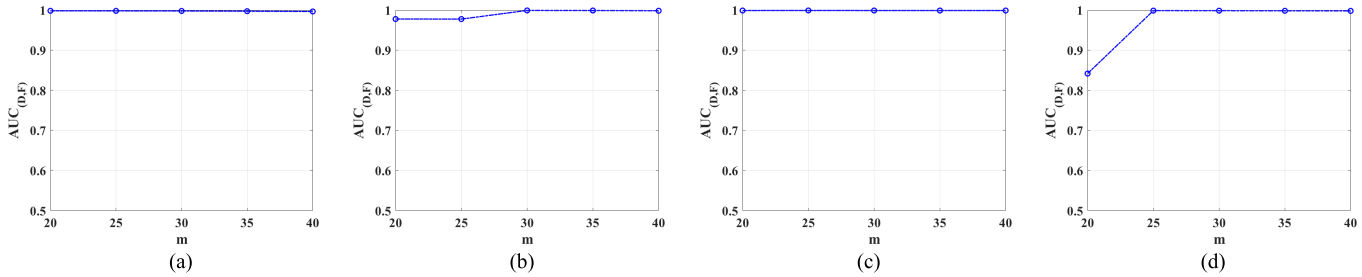


Fig. 7. Parameter analysis of mask number m on four datasets. (a) HYDICE Urban. (b) Pavia. (c) Hyperion. (d) Abu-beach-1.

strategy. Specifically, we compare the performance of the IGPAD and its three reduced versions which are depicted as follows.

1) *IGPAD:* The standard IGPAD.
2) *IGPAD-EM:* The standard IGPAD without the edge modification operations during the edge modification stage.
3) *IGPAD-IM:* The standard IGPAD which replaces the iterative mask strategy with direct mask strategy. In other words, all the suspicious vertices are masked in one turn.

Different from the extra modules in other models that are designed to improve detection performance, edge modification operations and iterative mask strategy are exploited to mainly improve the robustness of the IGPAD. Through the parameter analysis section which is mentioned above, there exists a big drop in performance if K and m are not suitable due to leaks of anomalous vertices during masking. To alleviate this problem, we utilize edge modification operations and iterative mask strategy to improve the probability of all anomalous vertices being masked under the same parameters K and m.

In order to better demonstrate the effectiveness of edge modification operations and iterative mask strategy against the anomalous vertices leak problem, we adjust parameters K and m on HYDICE Urban and Hyperion datasets. As shown in Table II, if the IGPAD lacks the ED or IM module, there is a possibility of a significant drop in the $AUC_{(D,F)}$ value as indicated by the downward arrow in Table II. The significant drop is caused by the anomalous vertices leak problem which means that some anomalous vertices are classified as suspectless vertices during the mask stage. With the EM and IM modules, anomalous vertices become more likely to be categorized as suspicious vertices and masked.

TABLE II
VALUES OF $AUC_{(D,F)}$ ON FOUR DATASETS FOR ABLATION STUDY OF EDGE MODIFICATION OPERATIONS AND ITERATIVE MASK STRATEGY

Dataset	IGPAD-EM	IGPAD-IM	IGPAD
HYDICE Urban	0.9026 ↓	0.9993	0.9991
Pavia	0.9992	0.9778 ↓	0.9992
Hyperion	0.9670 ↓	0.9673 ↓	0.9994
abu-beach-1	0.8327 ↓	0.9997	0.9997

E. Detection Performance

In this section, we explore the overall performance of the proposed IGPAD method both qualitatively and quantitatively. To demonstrate the superiority of IGPAD, the proposed IGPAD method is compared with other classical, graph-related or state-of-the-art methods over four small-size anomaly datasets. The qualitative comparisons are illustrated by detection maps, which give a visual intuition. And the quantitative performance is conveyed through nine metrics derived from 3-D ROC. Besides the figures and tables which give a visual representation of the detection performance, we also provide explicit descriptions and analysis of the detection performance over four datasets.

1) Performance Over HYDICE Urban Dataset: The qualitative detection performance over HYDICE Urban dataset is illustrated by Fig. 8. As we can see, most anomalies are highlighted in the detection map of the proposed IGPAD method while exhibiting terrific background suppression. For the GRX, CRD, GTVLRR, and LSMAD methods, although they enhance the most of anomalies in the detection map, the background of their detection maps is not clean enough. And the performance of background suppression is even worse in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

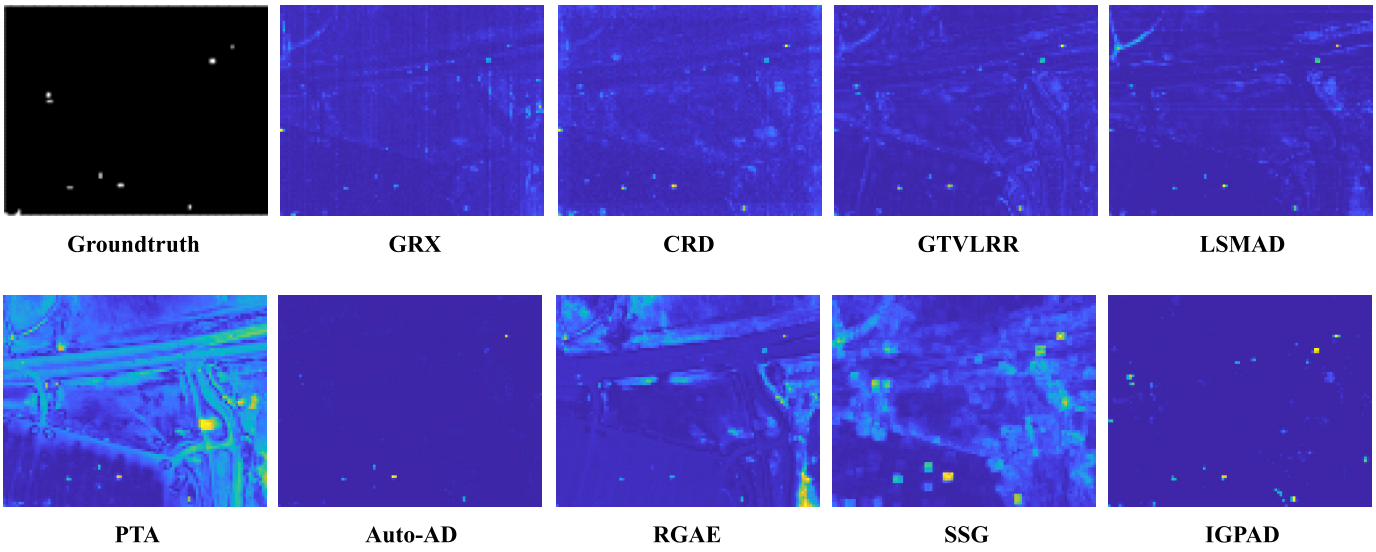


Fig. 8. Hyperspectral detection maps of comparison methods and proposed IGPAD method on HYDICE Urban dataset.

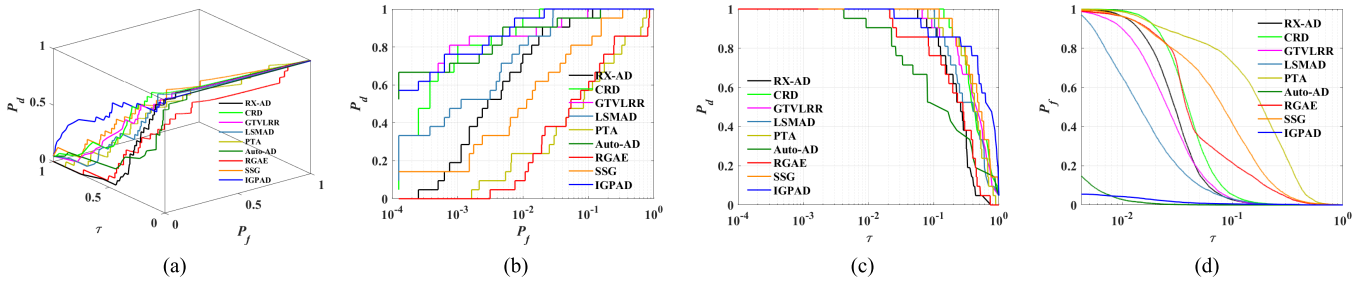


Fig. 9. ROC curves of nine anomaly detection methods over HYDICE Urban datasets. (a) 3-D ROC curve. (b) 2-D ROC curve of $(P_d - P_f)$. (c) 3-D ROC curve of $(P_d - \tau)$. (d) 3-D ROC curve of $(P_f - \tau)$.

TABLE III
VALUES OF AUC METRICS OF NINE METHODS OVER HYDICE URBAN DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TD-BS}	AUC_{ODP}
GRX	0.9872	0.2641	0.0361	1.2514	0.9511	7.3144	0.2280	1.2153
CRD	0.9978	0.4739	0.0472	1.4716	0.9506	10.0393	0.4267	1.4244
GTVLRR	0.9922	0.4541	0.0331	1.4463	0.9591	13.7004	0.4209	1.4131
LSMAD	0.9935	0.3878	0.0243	1.3813	0.9692	15.9807	0.3635	1.3570
PTA	0.8175	0.4478	0.2081	1.2653	0.6095	2.1518	0.2397	1.0572
Auto-AD	0.9899	0.2661	**0.0025**	1.2560	0.9874	**107.8364**	0.2637	1.2536
RGAE	0.8155	0.2741	0.0781	1.0895	0.7373	3.5092	0.1960	1.0114
SSG	0.9505	0.5070	0.1109	1.4575	0.8396	4.5709	0.3961	1.3466
IGPAD	**0.9979**	**0.6297**	0.0068	**1.6276**	**0.9911**	92.8820	**0.6229**	**1.6208**

the detection maps of the PTA, RGAE, and SSG methods which have extremely high rate of false alarm. Different from other comparison methods, Auto-AD performs well on background suppression, but some anomalies are also suppressed during detecting which leads to the problem of missing detection.

The 3-D ROC curve, 2-D ROC curve, as well as the derivative metrics are represented by Fig. 9 and Table III. The IGPAD method obtains the optimal results on $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TD-BS}, and AUC_{ODP}, as well as obtain suboptimal results on the rest metrics which demonstrate the superiority of the proposed IGPAD method on HYDICE Urban dataset.

2) Performance Over Pavia Dataset: The detection maps over Pavia dataset are shown in Fig. 10. There exist two major drawbacks in the comparison detection methods. One is the poor performance on background suppression which occurs in the detection map of the CRD, GTVLRR, PTA, RGAE, and SSG, and the other is that the anomalies are not sufficiently highlighted. In contrast, the IGPAD method acquires satisfactory results on both background suppression and target detectability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

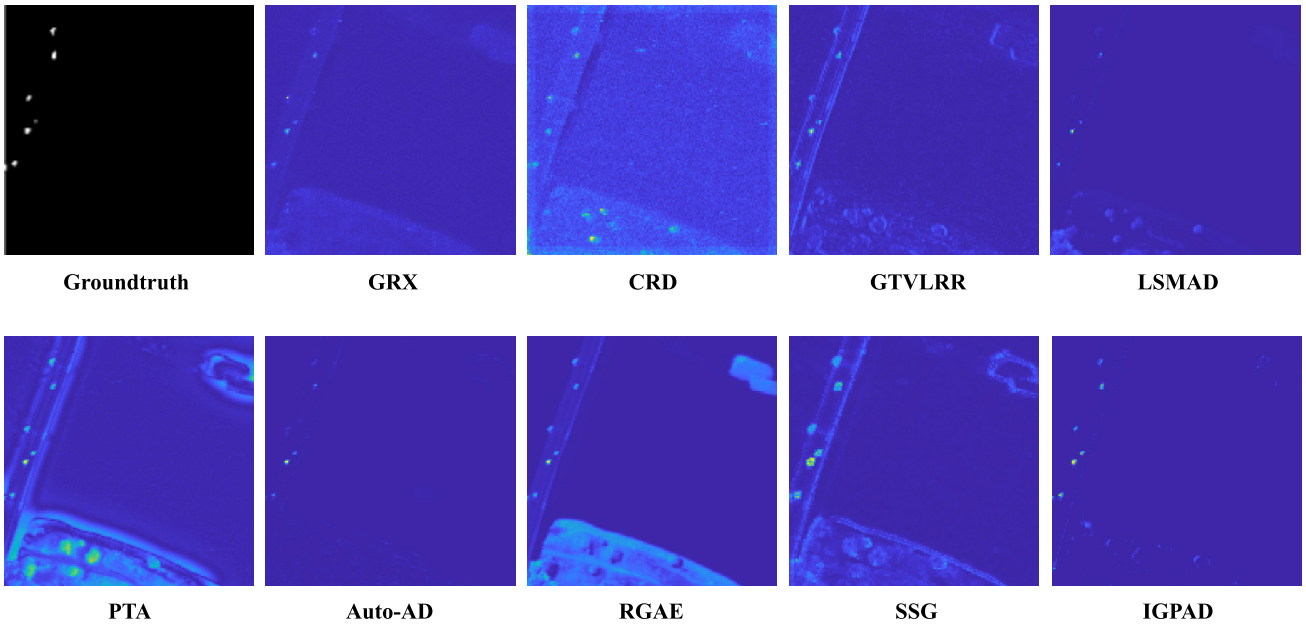


Fig. 10. Hyperspectral detection maps of comparison methods and proposed IGPAD method on Pavia dataset.

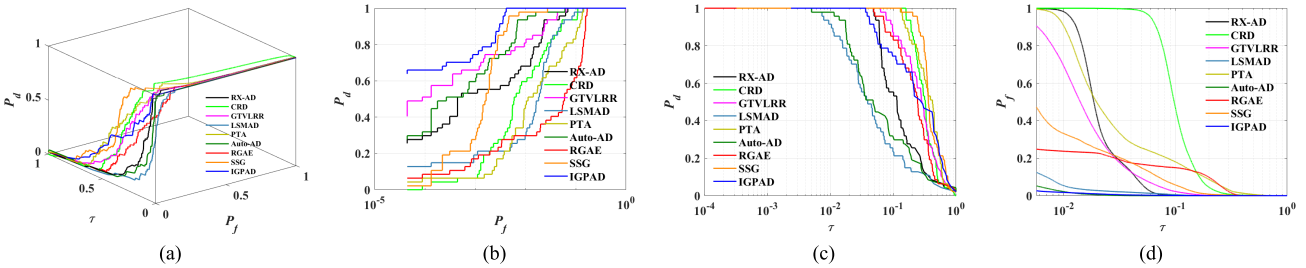


Fig. 11. ROC curves of nine anomaly detection methods over Pavia datasets. (a) 3-D ROC curve. (b) 2-D ROC curve of $(P_d - P_f)$. (c) 3-D ROC curve of $(P_d - \tau)$. (d) 3-D ROC curve of $(P_f - \tau)$.

TABLE IV
VALUES OF AUC METRICS OF NINE METHODS OVER PAVIA DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	$AUC_{TD\text{-}BS}$	AUC_{ODP}
GRX	0.9905	0.1730	0.0233	1.1635	0.9672	7.4149	0.1497	1.1402
CRD	0.9811	0.3487	0.1092	1.3298	0.8719	3.1922	0.2395	1.2206
GTVLRR	0.9935	0.3173	0.0222	1.3109	0.9714	14.3233	0.2952	1.2887
LSMAD	0.9763	0.1047	0.0034	1.0810	0.9729	31.2057	0.1014	1.0776
PTA	0.9635	0.3952	0.0590	1.3588	0.9045	6.7001	0.3362	1.2998
Auto-AD	0.9966	0.1364	**0.0013**	1.1329	**0.9953**	**104.3319**	0.1351	1.1316
RGAE	0.9277	0.2678	0.0380	1.1955	0.8896	7.0451	0.2298	1.1575
SSG	0.9976	**0.4217**	0.0230	**1.4192**	0.9746	18.3322	**0.3987**	**1.3962**
IGPAD	**0.9992**	0.3373	0.0047	1.3365	0.9944	71.1572	0.3326	1.3318

The ROC curves and the corresponding AUC metrics are shown in Fig. 11 and Table IV. The IGPAD method achieves the best or second best results on $AUC_{(D,F)}$, $AUC_{(F,\tau)}$, AUC_{BS}, AUC_{SNPR}, and AUC_{ODP}, which indicates the effectiveness of the proposed IGPAD method on Pavia dataset.

3) Performance Over Hyperion Dataset: Fig. 12 shows the detection maps over Hyperion dataset. The PTA and SSG methods fail to suppress the background; thus, their visualization maps are unsatisfactory. For the GRX, CRD, LSMAD, Auto-AD, and RGAED methods, they cannot guarantee efficient anomaly target detection and background suppression simultaneously while the proposed IGPAD method achieves a better balance between target detection and background suppression.

The 3-D ROC curves and 2-D ROC curves are displayed in Fig. 13, and the corresponding metrics are listed in Table V. On the three overall metrics of $AUC_{(D,F)}$, $AUC_{TD\text{-}BS}$, and AUC_{ODP}, the IGPAD method obtains optimal results. Additionally, the IGPAD method obtains the suboptimal results on AUC_{TD}, AUC_{BS}, and AUC_{SNPR}. The excellent AUC metric

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

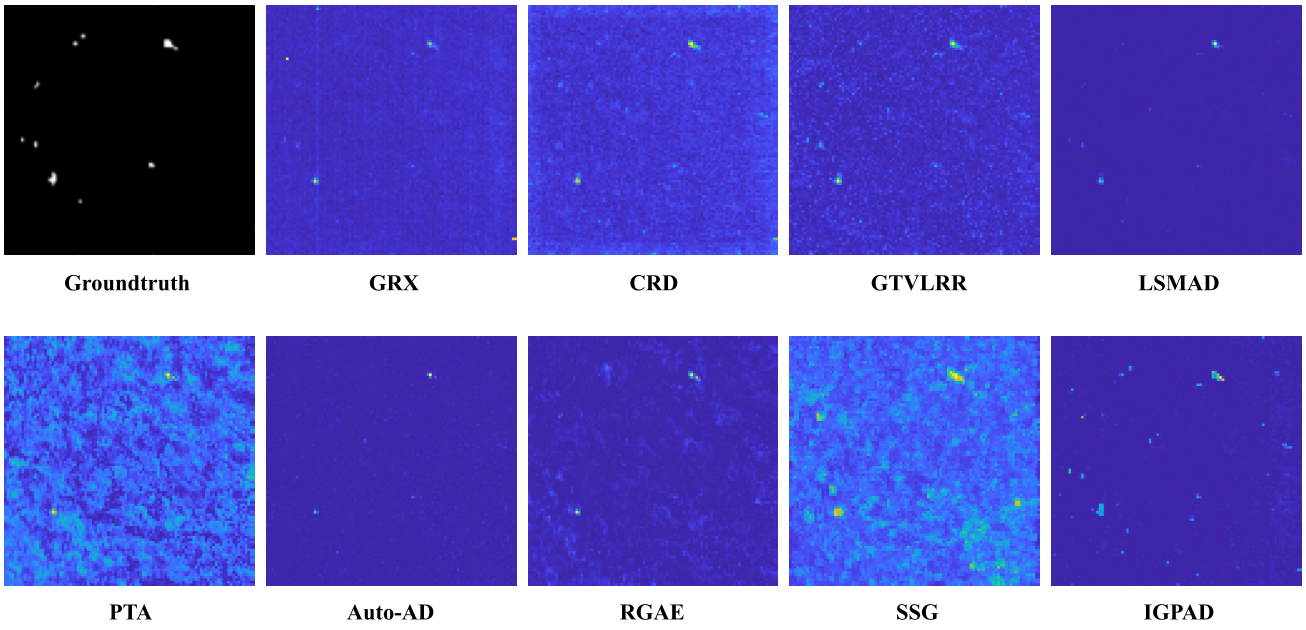


Fig. 12. Hyperspectral detection maps of comparison methods and proposed IGPAD method on Hyperion dataset.

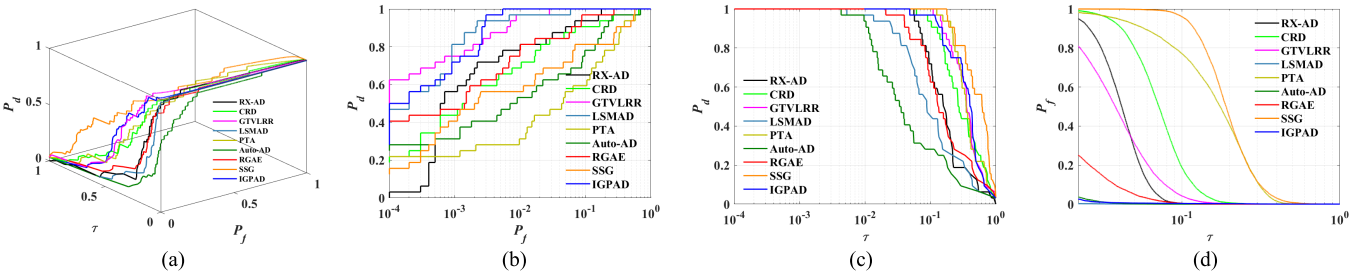


Fig. 13. ROC curves of nine anomaly detection methods over Hyperion datasets. (a) 3-D ROC curve. (b) 2-D ROC curve of $(P_d - P_f)$. (c) 3-D ROC curve of $(P_d - \tau)$. (d) 3-D ROC curve of $(P_f - \tau)$.

TABLE V
VALUES OF AUC METRICS OF NINE METHODS OVER HYPERION DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TD-BS}	AUC_{ODP}
GRX	0.9829	0.2319	0.0435	1.2148	0.9395	5.3358	0.1884	1.1714
CRD	0.9574	0.3678	0.0768	1.3252	0.8806	4.7866	0.2910	1.2484
GTVLRR	0.9980	0.3969	0.0440	1.3949	0.9540	9.0222	0.3529	1.3509
LSMAD	0.9974	0.1733	**0.0018**	1.1708	**0.9956**	**95.2073**	0.1715	1.1689
PTA	0.8676	0.4063	0.1802	1.2739	0.6874	2.2542	0.2260	1.0937
Auto-AD	0.9259	0.1114	0.0079	1.0373	0.9180	14.0818	0.1035	1.0294
RGAE	0.9795	0.2401	0.0166	1.2197	0.9630	14.4938	0.2236	1.2031
SSG	0.9106	**0.5689**	0.2060	**1.4795**	0.7046	2.7612	0.3629	1.2735
IGPAD	**0.9992**	0.4407	0.0121	1.4399	0.9872	36.5693	**0.4286**	**1.4278**

values validate the overall effectiveness of the IGPAD method on Hyperion dataset.

4) Performance Over Abu-Beach-1 Dataset: The detection maps over abu-beach-1 dataset are exhibited in Fig. 14. Although there is only a single anomaly object in the dataset, it is still challenging to separate the anomaly object from the beach which is located in the southwest of the image. As shown in Fig. 14, the eight comparison methods are unable to separate anomaly object from the beach well. Even worse, some methods like PTA and RGAE highlight the beach while suppress anomaly object. Conversely, the IGPAD method successfully separate anomaly object from the beach, and most of the beach are perfectly suppressed in the detection map.

Fig. 15 exhibits the ROC curves, and Table VI lists the corresponding metrics. On all the metrics except $AUC_{(F,\tau)}$, the IGPAD method obtains the optimal or suboptimal results. Although the $AUC_{(F,\tau)}$ value of IGPAD is not optimal or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

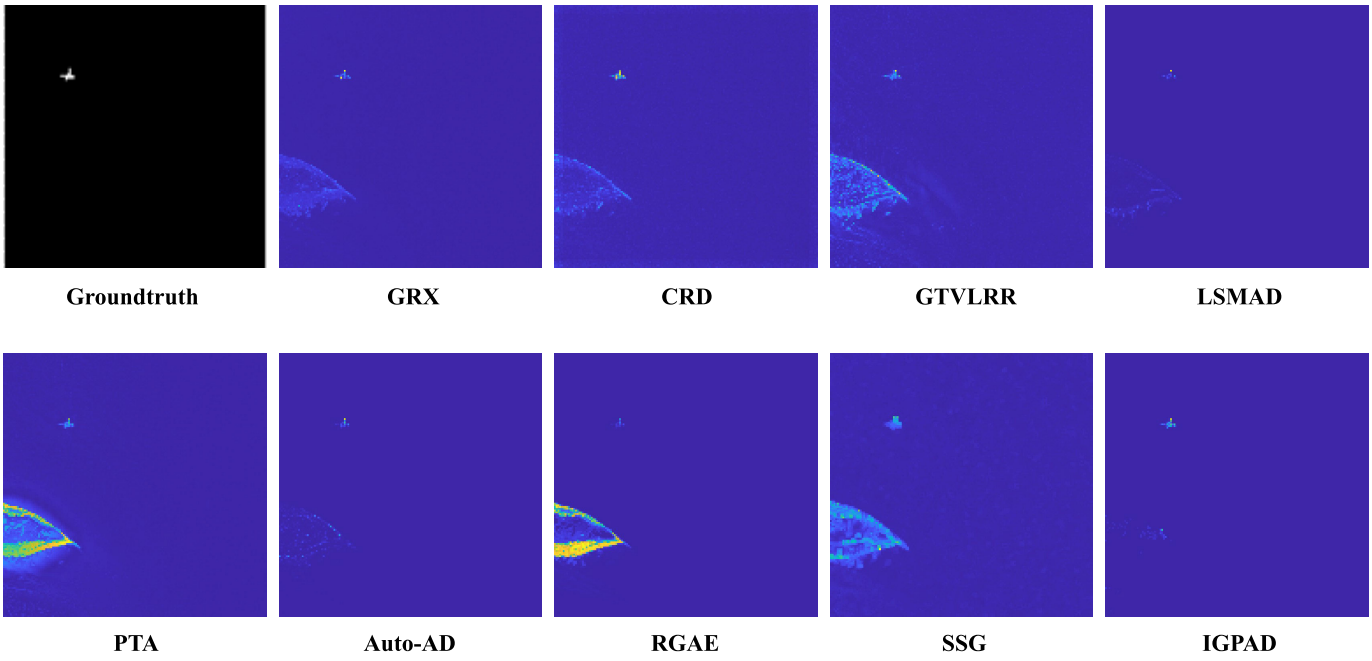


Fig. 14. Hyperspectral detection maps of comparison methods and proposed IGPAD method on abu-beach-1 dataset.

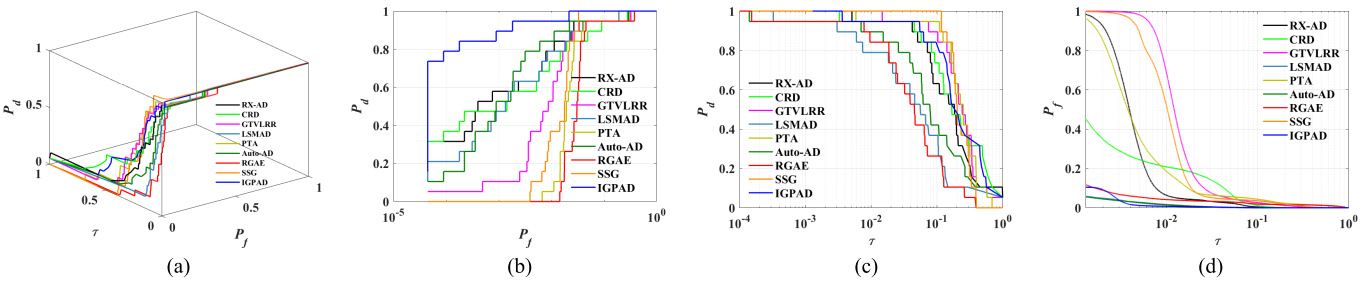


Fig. 15. ROC curves of nine anomaly detection methods over abu-beach-1 datasets. (a) 3-D ROC curve. (b) 2-D ROC curve of $(P_d - P_f)$. (c) 3-D ROC curve of $(P_d - \tau)$. (d) 3-D ROC curve of $(P_f - \tau)$.

TABLE VI
VALUES OF AUC METRICS OF NINE METHODS OVER ABU-BEACH-1 DATASET

Method	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TD-BS}	AUC_{ODP}
GRX	0.9796	0.2145	0.0066	1.1941	0.9731	32.7240	0.2079	1.1876
CRD	0.9735	0.2826	0.0108	1.2560	0.9627	26.1711	0.2718	1.2452
GTVLRR	0.9719	0.2453	0.0201	1.2172	0.9518	12.1938	0.2252	1.1971
LSMAD	0.9833	0.1099	**0.0007**	1.0932	0.9826	148.3710	0.1091	1.0925
PTA	0.9642	0.2759	0.0212	1.2401	0.9429	13.0003	0.2547	1.2189
Auto-AD	0.9809	0.1571	0.0008	1.1379	0.9800	**188.5695**	0.1562	1.1371
RGAE	0.9530	0.0788	0.0132	1.0318	0.9399	5.9728	0.0656	1.0186
SSG	0.9835	0.2351	0.0191	1.2186	0.9644	12.2955	0.2159	1.1995
IGPAD	**0.9997**	**0.3394**	0.0024	**1.3391**	**0.9973**	140.5955	**0.3370**	**1.3367**

suboptimal, the value is considerably low which demonstrates the powerful background suppression capabilities of IGPAD.

V. CONCLUSION

In this article, we propose a novel graph-based method called IGPAD which leverages the spectral correlation of pixels. To model the spectral correlation of pixels, we adopt a mutual-KNN strategy to construct the spectral-correlation graph whose edge weight is related to the spatial distance between two vertices. Besides, we apply EA and ED operations to modify the graph and improve the robustness of IGPAD. Then, we utilize the iterative mask strategy to mask all the anomalous vertices as much as possible. Finally, the iterative graph propagation update strategy is employed to realize the vertices reconstruction based on the information implied by the graph structure. To demonstrate the effectiveness of the edge modification and the iterative mask modules, we conduct an ablation study which proves the contribution of EM and IM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

modules to robustness. Moreover, comparison experiments of IGPAD with the same parameters on four datasets verify the robustness and superiority of the proposed IGPAD method. Although we have repeatedly emphasized that anomaly should be small size and the IGPAD is designed for small-size anomaly detection, we still believe that we need to strengthen the ability of IGPAD to detect relatively large anomalies which is one of the future directions for improvements in IGPAD.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[4] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[5] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[6] Chein-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[7] C.-I. Chang, "Effective anomaly space for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526624.

[8] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Aug. 1990.

[9] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[10] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[11] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[12] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016.

[13] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[14] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.

[15] Y. Qu et al., "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[16] X. Zhang, X. Ma, N. Huyan, J. Gu, X. Tang, and L. Jiao, "Spectral-difference low-rank representation learning for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10364–10377, Dec. 2021.

[17] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[18] Z. Wu, H. Su, and P. Zheng, "Hyperspectral anomaly detection using collaborative representation with PCA remove outlier," in *Proc. 9th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2018, pp. 1–5.

[19] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417.

[20] M. Imani, "Anomaly detection using morphology-based collaborative representation in hyperspectral imagery," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 457–471, Jan. 2018.

[21] R. Wang, H. Hu, F. He, F. Nie, S. Cai, and Z. Ming, "Self-weighted collaborative representation for hyperspectral anomaly detection," *Signal Process.*, vol. 177, Dec. 2020, Art. no. 107718.

[22] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[23] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[24] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[25] Q. Wang, J. Zeng, H. Wu, J. Wang, and K. Sun, "Self-adaptive low-rank and sparse decomposition for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3672–3685, 2022.

[26] X. Zhang, G. Wen, and W. Dai, "A tensor decomposition-based anomaly detection algorithm for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5801–5820, Oct. 2016.

[27] A. Ruhan, X. Mu, and J. He, "Enhance tensor RPCA-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] X. Zhang and G. Wen, "A fast and adaptive method for determining K_1, K_2, and K_3 in the tensor decomposition-based anomaly detection algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 3–7, Jan. 2018.

[29] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[30] S. Xiao, T. Zhang, Z. Xu, J. Qu, S. Hou, and W. Dong, "Anomaly detection of hyperspectral images based on transformer with spatial–spectral dual-window mask," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1414–1426, 2023.

[31] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[32] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.

[33] X. Lu, W. Zhang, and J. Huang, "Exploiting embedding manifold of autoencoders for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1527–1537, Nov. 2020.

[34] S. Arisoy, N. M. Nasrabadi, and K. Kayabol, "GAN-based hyperspectral anomaly detection," in *Proc. 28th Eur. Signal Process. Conf. (EUSIPCO)*, Jan. 2021, pp. 1891–1895.

[35] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[36] J. Lei, W. Xie, J. Yang, Y. Li, and C.-I. Chang, "Spectral–spatial feature extraction for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8131–8143, Oct. 2019.

[37] W. Xie, S. Fan, J. Qu, X. Wu, Y. Lu, and Q. Du, "Spectral distribution-aware estimation network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512312.

[38] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS^{3}LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[39] D. W. Messinger and J. Albano, "A graph theoretic approach to anomaly detection in hyperspectral imagery," in *Proc. 3rd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, 2011, pp. 1–4.

[40] F. Verdoja and M. Grangetto, "Graph Laplacian for image anomaly detection," *Mach. Vis. Appl.*, vol. 31, nos. 1–2, p. 11, Feb. 2020.

[41] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.

[42] X. Yang, B. Tu, Q. Li, J. Li, and A. Plaza, "Graph evolution-based vertex extraction for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Aug. 25, 2024, doi: 10.1109/TNNLS.2023.3303273.

[43] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[44] I. Rolland, S. Selvakumaran, and A. Marinoni, "Beyond low rank: A graph-based propagation approach to tensor completion for multi-acquisition scenarios," 2023, *arXiv:2312.03436*.

[45] Y. Zhang, G. Cao, B. Wang, X. Li, P. Y. O. Amoako, and A. Shafique, "Dual sparse representation graph-based copropagation for semisupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500517.

[46] B. Tu, Z. Wang, H. Ouyang, X. Yang, J. Li, and A. Plaza, "Hyperspectral anomaly detection using the spectral–spatial graph," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542814.

[47] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.



Jiahui Sheng received the B.S. degree from Zhejiang University, Hangzhou, China, in 2023, where he is currently pursuing the Ph.D. degree in control theory and control engineering.

His research interests include hyperspectral anomaly detection, image processing, and pattern recognition.



Xiaorun Li received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1992, and the M.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1995 and 2008, respectively.

Since 1995, he has been with Zhejiang University, where he is currently a Professor with the College of Electrical Engineering. His research interests include hyperspectral image processing, signal and image processing, and pattern recognition.



Shuhan Chen (Member, IEEE) received the B.S. degree from Ludong University, Yantai, China, in 2011, the M.S. degree from Liaoning Technical University, Huludao, China, in 2014, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2020.

She did research as a Visiting Faculty Assistant at the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, from 2018 to 2020. She did research as a Post-Doctoral Researcher at the Department of Control Science and Engineering, Zhejiang University, where she is currently an Assistant Professor. Her research interests include image registration, hyperspectral image processing, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:34:47 UTC from IEEE Xplore. Restrictions apply.